

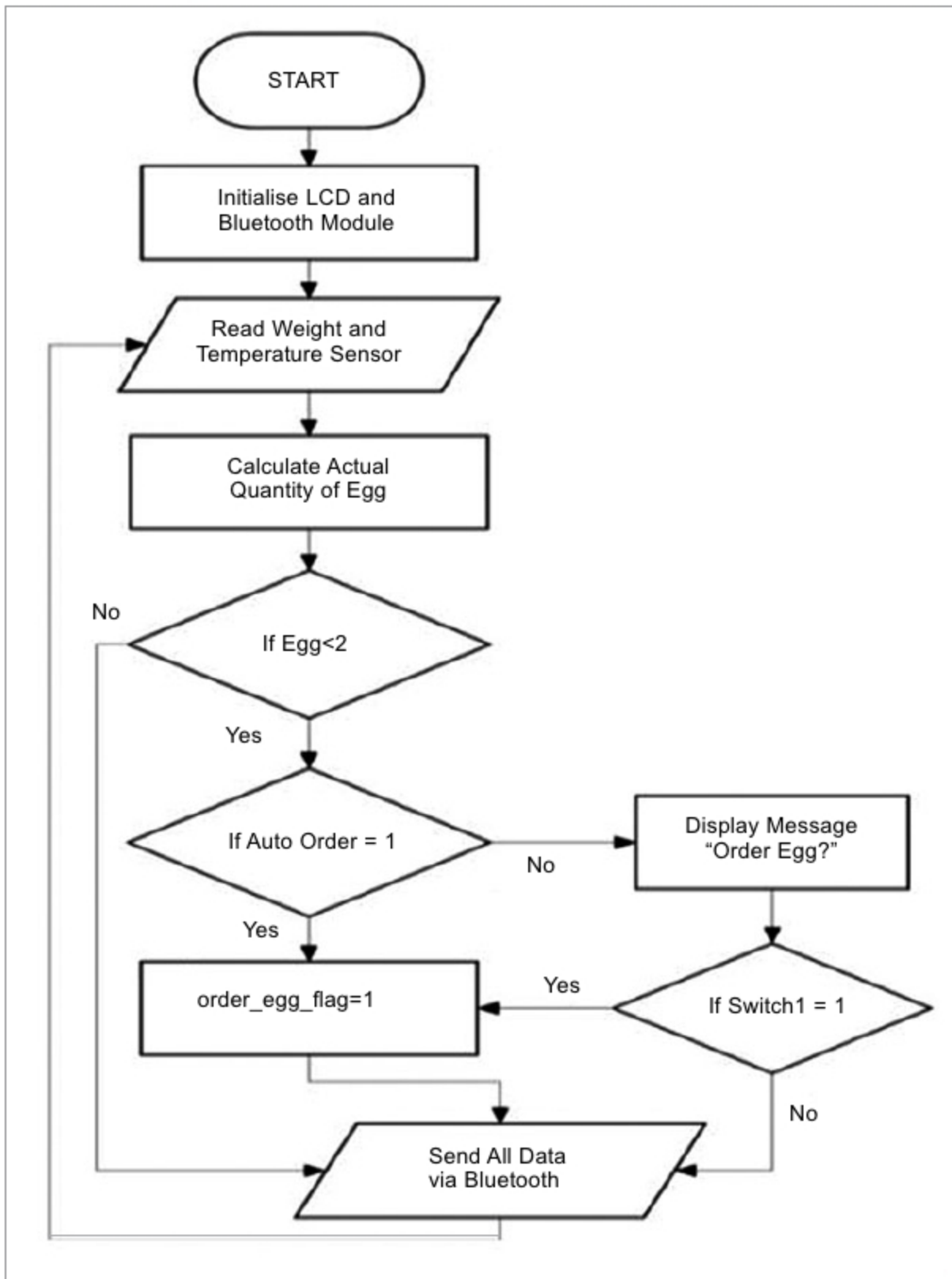


Fig. 2: The flowchart

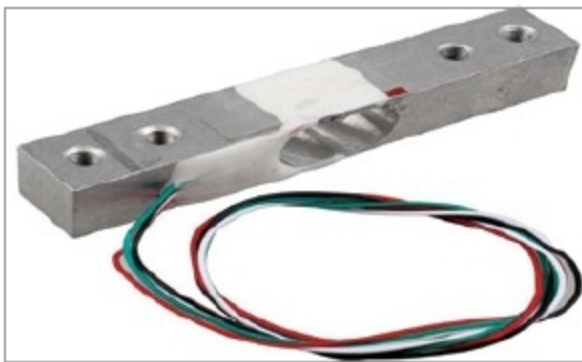


Fig. 3: Load cell



Fig. 4: Manual switches

Hardware used

The different sensors and hardware used in the system are as follows:

Raspberry Pi 3. Raspberry Pi 3 Model B has 1.2GHz quad-core ARM Cortex-A53 CPU and 1GB RAM. Like Pi 1 Model B+, it also has four USB ports, forty GPIO pins, full HDMI port, Ethernet port, combined 3.5mm audio jack and composite video, camera interface (CSI), display interface (DSI), micro-SD card slot, Video Core IV 3D graphics core as well as inbuilt Bluetooth and Wi-Fi. Raspberry Pi 3 board is



Fig. 5: Raspberry Pi 3 board

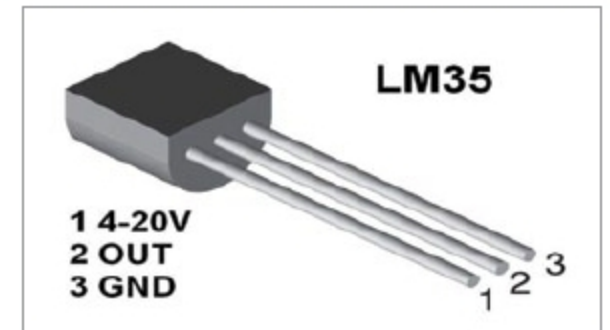


Fig. 6: LM35 temperature sensor

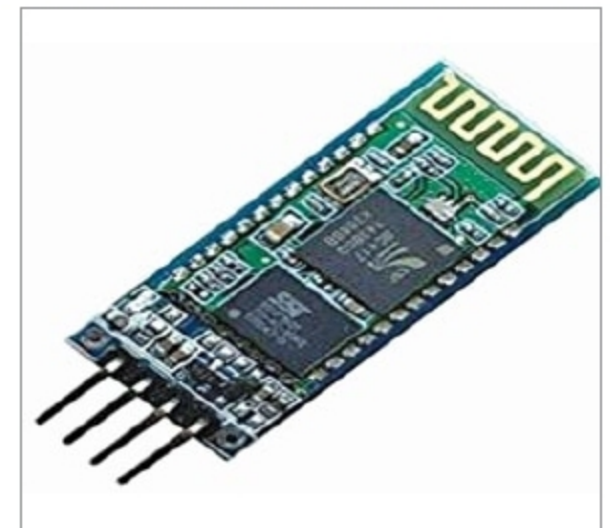


Fig. 7: Bluetooth module HC05

shown in Fig. 5.

Load cell (weight sensor). Load cell is an electronic transducer that generates an electrical signal depending on the force applied to it. YZC-131 load cell can measure up to three kilograms of groceries. It has sensitivity of $2 \pm 10\%$ (mV/V), input resistance of $415 \pm 15\Omega$ and an operating voltage of 10V DC. The weight sensor is shown in Fig. 3.

Temperature sensor. In this project LM35 temperature sensor is used for measuring temperature. It is shown in Fig. 6.

Bluetooth module. Bluetooth module transceiver is used for transmitting and/or receiving data. There are various Bluetooth modules that act as either master and/or slave. Bluetooth module HC05 is one such module. It uses Serial Port Protocol (SPP) to set up the wireless serial connection. It transmits values of